

The background features a complex network of thin, light-colored lines and dots, forming various triangular shapes and a larger, interconnected web-like structure. The lines are primarily in shades of light green and yellow, while the dots are small and light grey. The overall effect is a subtle, technical, and geometric aesthetic.

Timer 2 Initialization Overview

Overview & Implementation Steps

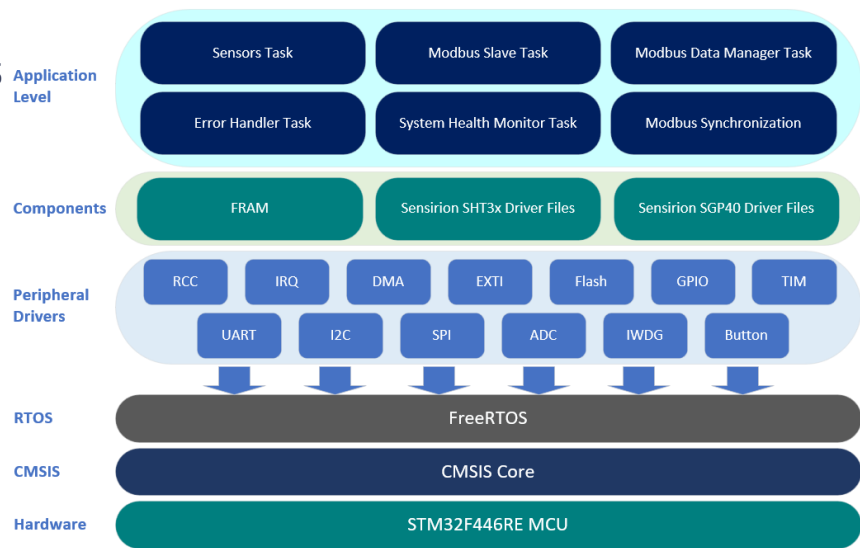
Timer 2 Initialization Overview

- Timers Used in the Project
- TIM2 Initialization Function

Timers Used in the Project

Timers General

- **Purpose & Functionality:** Crucial for generating precise delays, triggering periodic events, measuring time intervals and ensure synchronized operations across various tasks and peripherals.
- **Project Significance:**
 - **TIM2:** Triggers ADC1 conversions for the internal temperature sensor at 1 Hz.
 - **TIM3:** Generates high-frequency interrupts for FreeRTOS runtime statistics.
 - **TIM5:** Creates microsecond delays, required by the Sensirion drivers.



TIM2 Initialization Function

Overview: Timer 2 is a general-purpose 32-bit timer configured to trigger ADC1 conversions at a specific frequency.

- ✓ **TIM2 Clock Enable:** Using the `RCC_TIM2_CLK_ENABLE ()` macro, enable the TIM2 clock connected to the APB1 bus.

- ✓ **Prescaler & ARR:**

- ✓ **Prescaler:** Configures the timer frequency; APB1 clock at 90 MHz and a prescaler (PSC) of 8999, the timer frequency is 10 KHz.

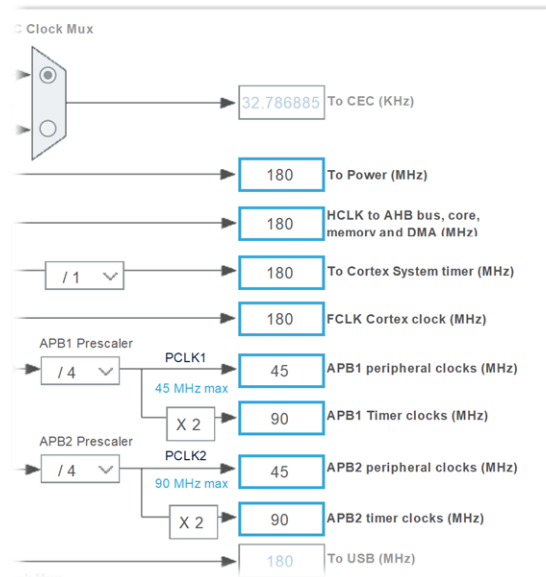
$$\text{Timer frequency} = 90 \text{ MHz} / (1 + 8999) = 10 \text{ KHz}$$

- ✓ **ARR:** Sets the trigger frequency to 1 Hz.

$$\text{Trigger frequency} = 10,000 \text{ Hz} / (1 + 9999) = 1 \text{ Hz}$$

- ✓ **Master Mode Selection:** Set the update event as the trigger output (TRGO) by configuring the MMS bits in CR2.
- ✓ **Event Generation:** Set the UG bit to generate an update event.
- ✓ **Counter Enable:** Enable the TIM2 counter to start the timer.

- **Project Significance:** Completing the timer setup for ADC1 trigger ensures the ADC performs conversions based on the external trigger from TIM2.



The background features a complex network of thin, light-colored lines and small circles, forming a web-like structure. Overlaid on this are several translucent triangles of various sizes and orientations, some of which are filled with a light yellow or green color. The overall aesthetic is technical and geometric.

Next: TIM Header File
